Computer Vision

EMARO- European Master on Advanced Robotics

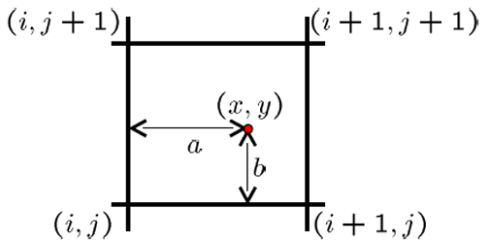
Robotics Engineering Master Degree

Image warping and bilinear interpolation

Given a coordinate transform (x,y)=h(x',y') and a source image f(x',y'), compute a transformed image g(x,y) = f(h(x',y')).

- Define a custom function for bilinear interpolation
- Perform backward warping (from the output image to the input image) on the provided image using the custom bilinear interpolation implementation
    - Transformations to implement:
        - Translation (try different translation parameters)
        - Rotation (try different angles)
        - Try your custom transformation

Bilinear Interpolation



$a = x - floor(x)$

$b = y - floor(y)$

$$f(x,y) = \begin{array}{ll} (1-a)(1-b) & f[i,j] \\ +a(1-b) & f[i+1,j] \\ +ab & f[i+1,j+1] \\ +(1-a)b & f[i,j+1] \end{array}$$